

Interfaces and Cables

Core Concepts & Standards

1. Introduction to Ethernet

Ethernet is a **collection** of network protocols and standards (not a single protocol), defined under **IEEE 802.3**. Standards ensure all vendors agree on physical and logical specifications so devices can communicate.

Network speed is measured in **bits per second (bps)**:

- 1 bit = a 0 or 1 (binary)
- 1 byte = 8 bits
- Units: Kilobit → Megabit → Gigabit → Terabit (each ×1,000)

Note: Hard drive storage uses *bytes*; network speed uses *bits*. A Gigabyte is 8× larger than a Gigabit.

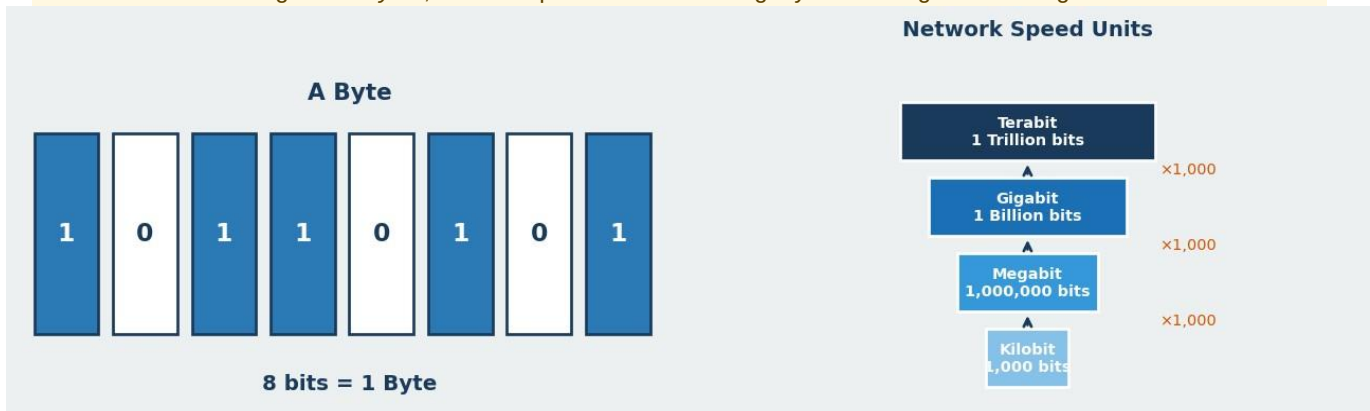


Figure 1 — Bit structure of a Byte (left) and Network Speed Units ladder (right)

2. Copper UTP Cable Standards

UTP = Unshielded Twisted Pair — 4 twisted pairs (8 wires total), terminated with **RJ-45** connectors (8 pins). Max cable length: **100 meters**.

Speed	Common Name	IEEE Standard	Informal Name	Wires Used
10 Mbps	Ethernet	802.3i	10BASE-T	2 pairs (4 wires)
100 Mbps	Fast Ethernet	802.3u	100BASE-TX	2 pairs (4 wires)
1 Gbps	Gigabit Ethernet	802.3ab	1000BASE-T	4 pairs (8 wires)
10 Gbps	10-Gig Ethernet	802.3an	10GBASE-T	4 pairs (8 wires)

- BASE = baseband signalling; T = twisted pair
- 1000BASE-T and 10GBASE-T: all 4 pairs are **bidirectional** (not dedicated Tx/Rx)

3. Pin Usage & Cable Types

Assignments for 10/100BASE-T:

Device	Transmit (Tx) Pins	Receive (Rx) Pins
PC / Router / Firewall	1, 2	3, 6
Switch	3, 6	1, 2

Note: Gigabit/10G also use pin pairs 4 & 5 and 7 & 8.

Full-Duplex: both devices send simultaneously using separate wire pairs — no collisions.

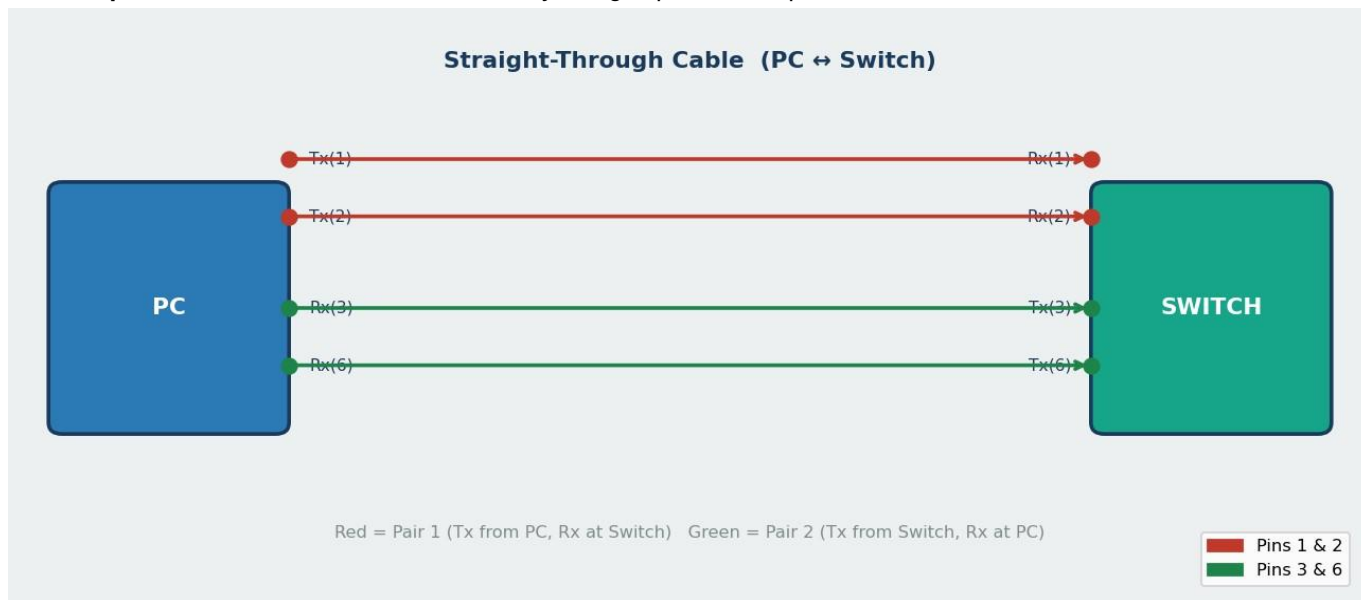


Figure 2 — Straight-Through Cable: PC ↔ Switch (unlike device types)

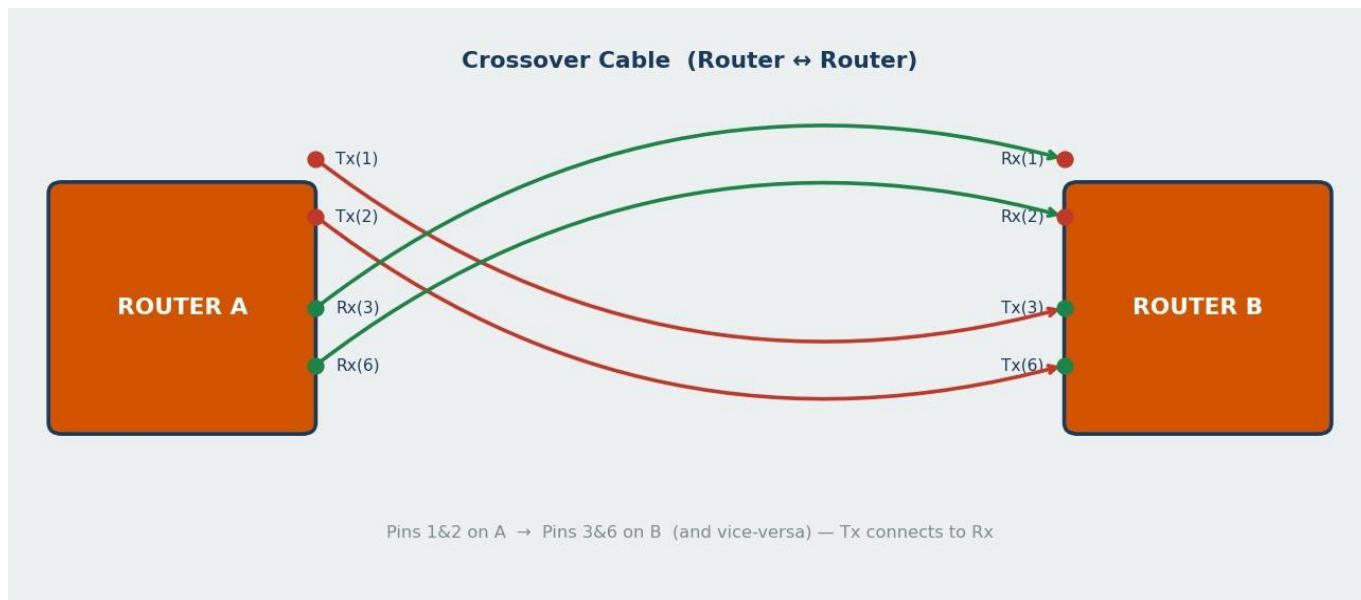


Figure 3 — Crossover Cable: Router ↔ Router (same device types — Tx crosses to Rx)

Auto MDI-X: Modern devices automatically detect and adjust pin usage — eliminates the need to choose cable type manually.

4. Fiber-Optic Cables

Used via **SFP transceivers** (Small Form-Factor Pluggable). Transmit data as **light over glass fiber**. Each end has two connectors — one Tx, one Rx.

Fiber-Optic Cable Cross-Section



Figure 4 — Cross-section of Multimode (left) vs Single-Mode (right) fiber optic cables

Feature	Multimode Fiber	Single-Mode Fiber
Core diameter	Wider	Narrower
Light source	LED-based (cheaper)	Laser-based (expensive)
Cable length	Longer than UTP; shorter than single-mode	Longest range
Cost	Cheaper	More expensive

Fiber-Optic Standards:

Standard	IEEE	Speed	Cable Type	Max Distance
1000BASE-LX	802.3z	1 Gbps	Multimode	550 m
1000BASE-LX	802.3z	1 Gbps	Single-mode	5 km
10GBASE-SR	802.3ae	10 Gbps	Multimode	400 m
10GBASE-LR	802.3ae	10 Gbps	Single-mode	10 km
10GBASE-ER	802.3ae	10 Gbps	Single-mode	30 km

Maximum Cable Distance Comparison

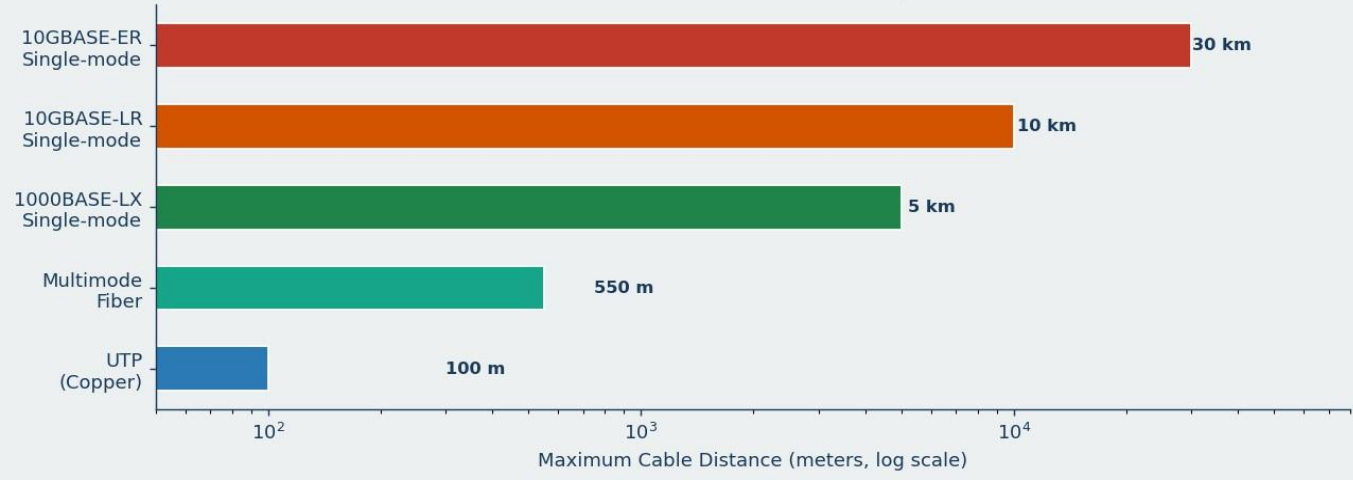


Figure 5 — Maximum cable distance comparison across all cable types (log scale)

5. UTP vs. Fiber-Optic Comparison

Feature	Copper UTP	Fiber-Optic
Cost	Cheaper	More expensive
Max distance	~100 m	Much longer
EMI vulnerability	Yes (twisting helps)	No
Port type	RJ-45 (cheaper)	SFP (more expensive)
Signal leakage / security	Yes — faint signal leaks	No risk

Key Terms

RJ-45 — Registered Jack 45; 8-pin connector for copper Ethernet

UTP — Unshielded Twisted Pair

SFP — Small Form-Factor Pluggable (fiber transceiver)

EMI — Electromagnetic Interference

Full-Duplex — Simultaneous two-way transmission

Auto MDI-X — Automatic pin detection/adjustment on modern devices

IEEE 802.3 — The Ethernet standards body

Quiz Questions & Answers

Q1: Two old routers with a UTP cable — data not sent/received. What is the problem?

Answer: A — Straight-through cable.

Option	Verdict	Reason
A. Straight-through cable	■ Correct	Both routers Tx on pins 1&2; straight connects Tx→Tx
B. Crossover cable	■ Incorrect	A crossover would actually fix the issue
C. Auto MDI-X mode	■ Incorrect	Auto MDI-X would fix it; old routers may lack it

Note: Same-type devices need a crossover cable when Auto MDI-X is unavailable.

Q2: Connecting switches ~150 m apart. Keep costs down. What cable?

Answer: C — Multimode fiber.

Option	Verdict	Reason
A. UTP	■ Incorrect	Max 100 m — too short
B. Single-mode fiber	■ Incorrect	Works but more expensive than needed
C. Multimode fiber	■ Correct	Covers 150+ m at lower cost than single-mode

Q3: Connecting two offices ~3 km apart. Keep costs down if possible. What cable?

Answer: B — Single-mode fiber.

Option	Verdict	Reason
A. UTP	■ Incorrect	Cannot support 3 km
B. Single-mode fiber	■ Correct	Many SM standards support well beyond 3 km
C. Multimode fiber	■ Incorrect	Cheaper but does not cover 3 km

Q4: Two identical switches with Auto MDI-X ports connected via straight-through cable. What happens?

Answer: A — They operate normally.

Option	Verdict	Reason
A. Operate normally	■ Correct	Auto MDI-X auto-adjusts — cable type irrelevant
B. Reduced speed	■ Incorrect	Without Auto MDI-X it would be zero, not reduced
C. Unable to communicate	■ Incorrect	Auto MDI-X resolves the issue

Note: Auto MDI-X makes straight-through vs. crossover cables irrelevant on modern devices.

Q5: Connecting many end hosts to a switch on the same office floor. What cable?

Answer: A — UTP.

Option	Verdict	Reason
A. UTP	■ Correct	Standard for hosts; switches have many RJ-45 ports
B. Single-mode fiber	■ Incorrect	Most hosts lack fiber NICs; not enough SFP ports
C. Multimode fiber	■ Incorrect	Same issue as single-mode

Cable Selection Decision Guide

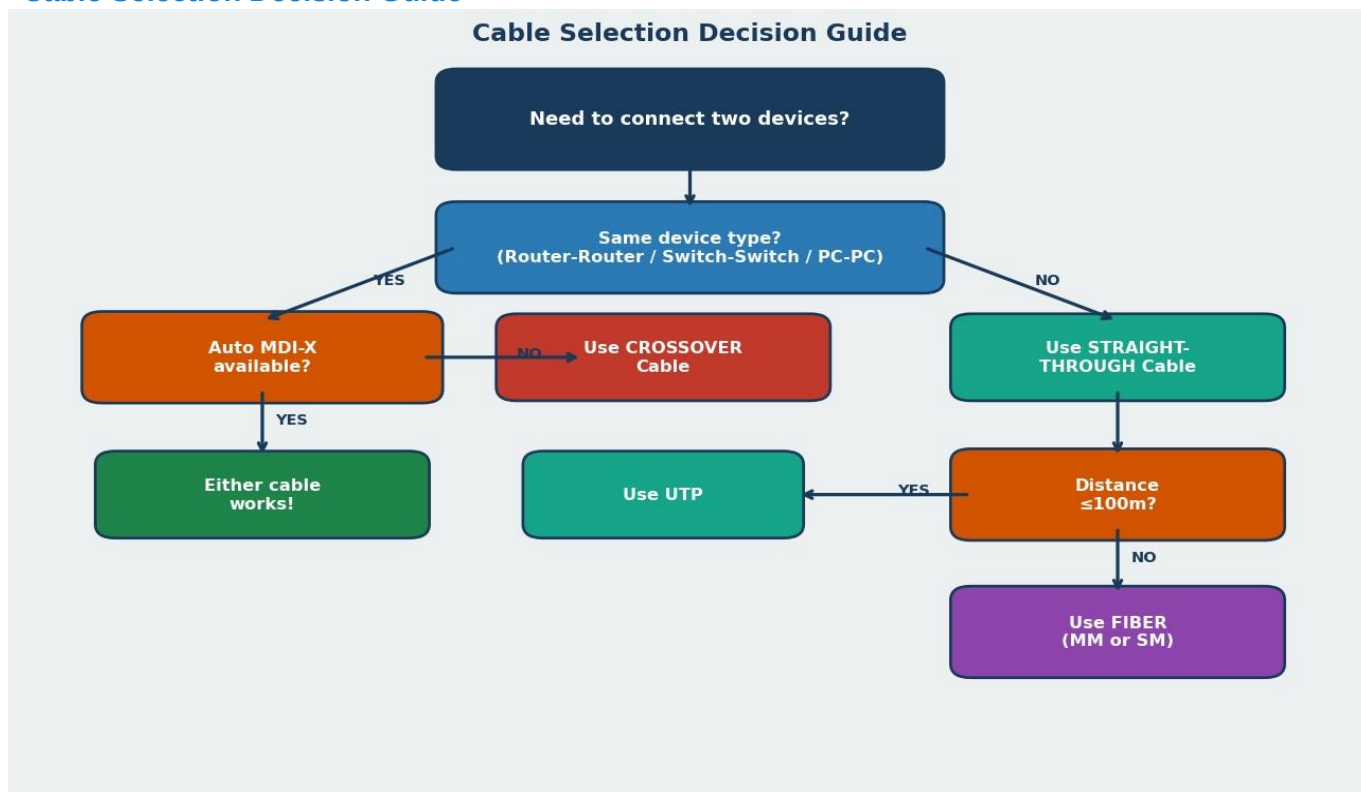


Figure 6 — Cable selection flowchart for common network scenarios